

MANISH KUMAR

Data-Driven Strategy & Sustainability Professional | Engineering · Analytics · Consulting
Boston, Massachusetts • F-1 / STEM-OPT Eligible • [manishkr2106.github.io](https://github.com/manishkr2106) • LinkedIn • GitHub: MK4Babson

PROFESSIONAL SUMMARY

Strategy and analytics professional combining 7+ years of civil/infrastructure engineering and project management with graduate-level business analytics and machine learning. Track record spanning World Bank-funded flood-control, water-resource and smart-city programs in India, and a Babson MBA focused on Business Analytics & ML (STEM). Bridges large-scale infrastructure delivery, data-driven decision modeling, and hands-on AI agent development. Seeks roles across management consulting, business/data analytics, sustainability strategy, and program management where engineering rigor, quantitative modeling, and AI tooling create measurable impact.

AREAS OF EXPERTISE

Consulting & Strategy: Strategy & operations, value-net / PARTS game-theory analysis, market & competitive analysis, business case development, stakeholder management, change management, client deliverables (decks, reports, Q&A prep)

Business & Data Analytics: Predictive modeling, classification & forecasting, dashboarding & BI, optimization, NLP / text analytics, A/B reasoning, KPI design, data storytelling

Project & Program Management: End-to-end project delivery, multi-stakeholder coordination, budgeting & scheduling, vendor/contractor management, World Bank-funded program execution, risk & quality control

Supply Chain & Operations: Network optimization (MILP), EOQ / newsvendor inventory models, reverse logistics, distribution planning, process improvement

Sustainability & Clean Energy: Climate-systems modeling, renewable-energy strategy, ESG / sustainability analytics, circular-economy initiatives

PROFESSIONAL EXPERIENCE

Project Manager

2017 – 2020 (≈3 yrs)

Water Resources / Smart City Infrastructure Programs

India

- **World Bank-funded delivery:** Managed flood-control, water-resource and smart-city infrastructure projects funded under World Bank programs, owning scope, schedule, budget and quality from planning through commissioning.
- **Stakeholder & vendor management:** Coordinated across government bodies, consultants, contractors and funding agencies; aligned multiple stakeholders on milestones, compliance and reporting requirements.
- **Budgeting & scheduling:** Built and tracked project budgets and timelines, managed contractor performance, and controlled cost and schedule variance across concurrent works.
- **Risk, quality & change control:** Implemented quality-assurance and risk-mitigation processes for large public-infrastructure works, applying engineering and change-management discipline now central to a sustainability-leadership focus.
- **Technical execution:** Used AutoCAD, HEC-RAS and GIS for hydraulic/flood modeling, design review and geospatial planning of water and urban infrastructure.

Assistant Engineer – Water Resources

2013 – 2017 (≈4 yrs)

Water Resources Department

India

- **Water-resource engineering:** Served as assistant engineer on water-resource and flood-management works, supporting design, estimation and field execution of hydraulic infrastructure.
- **Hydraulic modeling & design:** Applied HEC-RAS for hydraulic/flood-routing analysis and AutoCAD/GIS for design and geospatial assessment of channels, embankments and water-control structures.
- **Site supervision & quality:** Supervised construction quality, measurements and contractor deliverables; ensured compliance with engineering standards and public-works specifications.

- **Foundation for analytics career:** Built deep domain expertise in water systems, flood control and public-infrastructure delivery — the engineering backbone later integrated with MBA analytics and ML.

EDUCATION

Master of Business Administration (MBA) — Business Analytics & Machine Learning (STEM)

Expected May 2026

F.W. Olin Graduate School of Business, Babson College

Wellesley, MA

- Concentration in Business Analytics & Machine Learning (STEM-designated).
- **Core quantitative coursework:** Systems Thinking & Modeling (Vensim, causal-loop diagrams), International Macroeconomics, Economic & Financial Forecasting (ARIMA), Operations & Information Management (SQL / Python / SQLite), Data Mining & Machine Learning, Negotiation, Game Theory & Strategy.
- **ML methods applied:** kNN, Naïve Bayes, Logistic Regression, Neural Networks, Random Forest, CART, GBM, stacking ensembles, ARIMA, MILP network optimization.
- **Experiential programs:** Beta-X innovation lab; Babson Consulting Experience (BCE) with ChopValue Boston; active member, Babson Sustainability & Energy Club.

Prior Education — Bachelor's in Civil Engineering (India)

- Civil engineering degree underpinning 7+ years of water-resource, flood-control and infrastructure practice in India.

SELECTED PROJECTS — ANALYTICS, CONSULTING, AI & SUPPLY CHAIN

Artificial Intelligence & AI Agents

- **Babson Library Research Agent / Primo Agent** (*IBM Watsonx Orchestrate, Node.js MCP, Railway, SSE*): Built and deployed an AI library-research agent for Babson's Horn Library; Node.js MCP server (repo MK4Babson/babson-primo-mcp) on Railway connected to Watson Orchestrate via SSE, exposing three auto-discovered tools (search_library, search_by_type, get_resource_detail) over the Primo API. Part of an IBM Watsonx Orchestrate vs. Microsoft Copilot Studio comparison; delivered implementation report, presentation, Q&A prep and IT handover.
- **Primo Agent — Watsonx.ai version** (*IBM watsonx.ai, IBM Granite 3 8B, IAM / REST API, HTML/JS*): Standalone HTML chatbot integrating IBM IAM token exchange and the watsonx.ai text-generation endpoint (Granite 3 8B Instruct); structured prompt format returning JSON rendered as styled library-resource cards.

Machine Learning & Predictive Analytics

- **Boston Housing ML Classification** (*R: CART, Random Forest, GBM, stacking + NN meta-learner*): Binary classification of high-value properties using bagging, boosting and a stacked ensemble with a neural-network meta-learner; engineered factor conversion and probability extraction for the stacking layer.
- **Credit Default Prediction** (*Stacking ensemble (Python/R)*): Built a stacking-ensemble model to predict credit defaults, combining multiple base learners for improved classification performance.
- **Musk Tweet NLP Analysis** (*BERTopic, TextBlob, IBM Watsonx*): Topic modeling and sentiment analysis on a large tweet corpus, surfacing themes and sentiment trends.
- **Economic & Financial Forecasting** (*ARIMA, R*): Time-series forecasting of economic and financial indicators using ARIMA modeling and diagnostics.

Consulting & Strategy

- **ChopValue Boston — Babson Consulting Experience** (*BCE client engagement*): Live consulting engagement delivering market and operational recommendations for a circular-economy (recycled-chopstick) client; produced client-facing analysis and deliverables.
- **Peacock Streaming Strategy** (*Brandenburger & Nalebuff Value Net / PARTS, game theory*): Analyzed whether Peacock should compete on aggressive pricing vs. exclusive sports/media partnerships; built simultaneous-move payoff matrix, sequential game with rollback, and a plain-language recommendation.

- **GECA Climate Systems Model** (*Vensim, system dynamics*): Causal-loop and stock-and-flow model of a climate/sustainability system to test policy and intervention scenarios.

Supply Chain & Operations Optimization

- **Zara NYC Reverse Logistics & Donation Optimization** (*MILP, PuLP; EOQ / newsvendor; blockchain concept*): Mixed-integer optimization for reverse-logistics and donation routing; applied EOQ and newsvendor inventory models and explored blockchain-based traceability.
- **EcoGoods / Summit Maritime Distribution** (*Network optimization*): Distribution and network-optimization modeling for supply-chain efficiency across nodes and routes.

Business Intelligence & Data Engineering

- **RentSmart Boston** (*Tableau*): Interactive dashboard analyzing Boston housing-violation data to surface neighborhood-level risk and rental-quality insights.
- **BabsonAI Labs Database Design** (*SQLite, SQL*): Relational database design and implementation supporting an analytics-lab data model.
- **Python Analytics Pipeline** (*Python (pandas, matplotlib), Apache Airflow*): End-to-end analytics pipeline; built a real Apache Airflow DAG to implement and demonstrate workflow orchestration.

Innovation & Sustainability Concepts

- **Beta-X Innovation Lab** (*Product innovation*): Daily-wellness product concept and a dual-tip shoe-cleaner pen developed through Babson's innovation lab.
- **Renewable & Photonic Energy Concepts** (*Sustainability / clean-energy R&D*): Conceptual exploration of photonic energy transfer and wireless solar power, aligned with a clean-energy leadership focus.

TECHNICAL SKILLS & TOOLS

Programming & Data: Python (pandas, matplotlib, NetworkX, PuLP), R (caret, nnet, randomForest, gbm), SQL, SQLite

ML / Analytics: Logistic Regression, kNN, Naïve Bayes, Neural Networks, CART, Random Forest, GBM, Stacking Ensembles, ARIMA, BERTopic, TextBlob

Optimization: MILP, Gurobi, PuLP, EOQ / newsvendor models, network optimization

AI / GenAI: IBM Watsonx.ai, IBM Watsonx Orchestrate, IBM Granite, MCP servers, REST / IAM API integration, prompt engineering

BI & Visualization: Tableau, dashboarding, data storytelling

Modeling & Simulation: Vensim (system dynamics / causal loops), game-theory modeling

Workflow & Data Eng.: Apache Airflow (DAG orchestration), data pipelines

Engineering Tools: AutoCAD, HEC-RAS, GIS

Frameworks & Methods: Value Net / PARTS, Brandenburger-Nalebuff game theory, change management, World Bank program delivery

CAREER FOCUS & TARGET SECTORS

- **Management Consulting & Strategy:** Strategy & operations, sustainability and transformation advisory.
- **Business / Data Analytics:** BI analyst, data analyst and analytics-strategy roles.
- **Sustainability & Clean Energy:** Sustainability analyst and clean-energy strategy roles.
- **Program / Project Management:** Infrastructure, energy and technology program delivery.